

Muhammmad Athar Abbas
SP25-BSE-082
Object Oriented Programming
Lab Assignment 3

Task 1:

Shape Interface:

```
interface Shape{  
    double area();  
}
```

Task 2:

Circle Class:

```
public class Circle implements Shape{  
    double radius;  
    public Circle(double r){  
        radius = r;  
    }  
    @Override  
    public double area(){  
        double area = Math.PI* (radius * radius);  
        return area;  
    }  
}
```

Rectangle Class:

```
public class Rectangle implements Shape{  
    private double length;  
    private double width;  
    public Rectangle(double l, double w){
```

```
        length = l;
        width = w;
    }
    @Override
    public double area() {
        double area = length * width;
        return area;
    }
}
```

Triangle Class:

```
public class Triangle implements Shape {
    private double base;
    private double hyp;
    private double perp;
    public Triangle(double b, double h, double p) {
        base = b;
        hyp = h;
        perp = p;
    }
    @Override
    public double area() {
        double area = 0.5 * (base + perp); //for right angles.
        return area;
    }
}
```

Task 3:

Calculate Area Class:

```
public class CalculateAreas {
    public static double[] areaArray(Shape[] shapes) {
        double[] areas = new double[shapes.length];
    }
}
```

```

        for(int i = 0 ; i < shapes.length; i++){
            areas[i] = shapes[i].area();
        }
        return areas;
    }
}

```

TestClass:

```

public class TestClass{
    public static void main(String[] args) {
        Rectangle r = new Rectangle(10,20);
        System.out.println("Rectangle Area: " + r.area());
        Triangle t = new Triangle(10,20,30);
        System.out.println("Triangle Area: " + t.area());
        Circle c = new Circle(4.5);
        System.out.println("Circle Area: " + c.area());

        Shape[] shapeArray = {r,t,c};
        double[] areas = CalculateAreas.areaArray(shapeArray);
        System.out.println("All areas: [ " + areas[0] + ", " +
areas[1] + ", " + areas[2] + " ]");
    }
}

```

Output:

```

● [athar@athar ~/semester2/OOP/java_programs/drivelabs/lab9/tasks]$ java TestClass.java
Rectangle Area: 200.0
Triangle Area: 20.0
Circle Area: 63.61725123519331
All areas: [ 200.0, 20.0, 63.61725123519331 ]

```

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